Birudu Asish Mourya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.71
Bhashyam High School	Class XII (Board of Intermediate Education, Andhra Pradesh)	2023	98.20%
Bhashyam High School	Class X (Board of Secondary Education, Andhra Pradesh)	2021	100.00%

EXPERIENCE

Software Engineer Intern -- VividMinds Technologies Pvt Ltd

- Worked on a **tenant-configuration** based enterprise platform, using **C#** and **.NET** framework and shipping features across an **Angular** front end, .NET backend services and a **PostgreSQL** data layer shared across tenants.
- Implemented **role-based access control** spanning roles, permissions and tenant-scoped authorization, and extended the mailing list and templated email notification workflows used for invites, approvals and system alerts.

Backend Developer Intern -- mynoted.com

- Designed, developed and maintained scalable backend APIs with **Fastify** and **Node.js**, exposing **RESTful** endpoints and real-time functionality over **WebSockets** across production workloads and third-party client integrations.
- Integrated third-party services to extend platform capability and optimized backend processes for measurable improvements in performance, reliability and long-term maintainability of the service codebase and its downstream integrations.

PROJECTS

RDMA Congestion Control -- Hardware-Offloaded CCA Performance Analysis

- Studied and analysed the performance of hardware-offloaded CCAs (**DCQCN, SWIFT, EQDS**) on **RoCEv2** networks for research, evaluating reaction time, throughput recovery and signal-driven rate adaptation at sub-µs granularity under varied congestion, incast and background traffic patterns driven across the full testbed path at line rate.
- Set up a full **RDMA** testbed with **Marvell DAO** on an **Octeon CN10K DPU** and **DPDK** on a 100 GbE server, then implemented EQDS alongside DCQCN within DAO's RDMA stack to benchmark both under identical conditions.

VulnTracker -- Common Vulnerabilities & Exposures Tracker

- Developed a **Django** web application that ingests and normalizes real-time CVE data from multiple upstream security feeds into a single queryable vulnerability database backed by scheduled ingestion and de-duplication jobs.
- Built triage dashboards and advanced search across **CVSS** severity scoring, **CWE** classification, affected-product filters and detailed per-advisory reporting to support prioritized triage and remediation across large advisory sets.

Online Auction Platform

- Developed a **Django** auction platform with dynamic item listings, timed lots and real-time bidding.
- Led backend development, enforcing correct bid ordering and data integrity under concurrent updates alongside secure session-based authentication and per-user access control over listings, bids and sensitive account actions.

RISC-V Instruction Set Simulator

- Developed a **RISC-V** instruction set simulator in **C++** with a configurable cache hierarchy and main memory model, reporting hit and miss statistics across varied instruction mixes and cache configurations for analysis.
- Built an assembly input interface that visualizes register file updates and memory state at each cycle.

OTHER PROJECTS

- **Infinite Precision Arithmetic Library :** Developed a C++ lib for arithmetic operations on floating-point and integer types
- **1:10 Scale V8 Engine Model:** modeled a detailed V8 engine in Solid Edge CAD and 3D printed the individual components into a working assembly with realistic valvetrain and piston motion through the full cycle.
- **Demo Game:** built a short playable demo game in Unreal Engine 5 using Blueprints and built-in assets.

SKILLS

- **Languages :** C, C++, C#, Python, TypeScript, SQL, Assembly
- **Frameworks :** .NET, ASP.NET Core, Angular, React, Django, FastAPI, Fastify, Node.js
- **Databases :** PostgreSQL, MySQL
- **Tools :** Git, Docker, Postman, DPDK, Solid Edge, LATEX

RELEVANT COURSEWORK

- Computer Architecture, Operating Systems, Computer Networks, Compilers, Distributed AI Training in a Large-Scale GPU Cluster
- Machine Learning, Deep Learning, Multi-Agent Systems, Fraud Analytics using Predictive and Social Network Techniques
- Data Structures & Algorithms, Database Management Systems, Software Engineering, Linear Optimization, Practical Malware Analysis